

Steps to Build Your DBMS Project

Simple development plan for the project estimation system (Customer, Project, Room, Material, Estimate, Payment, Employee).

Step 1: Fix your SQL file

- Delete the stray word "workers" in Project_Employee
- Change "use sys;" to "use project;"
- Add one more table to connect Room and Material (your ER diagram has "material used", but the SQL doesn't)

Step 2: Create the database

- Run your SQL file in MySQL
- Check that all tables are created (show tables;)

Step 3: Add sample data

- Insert a few rows in each table (5 customers, 5 projects, 10 rooms, 10 materials, etc.)
- This helps you test everything later

Step 4: Write important SQL

- Queries: join tables, show project with customer name, total cost, etc.
- Triggers/procedures: only if your teacher asks (e.g. auto-calculate estimate)

Step 5: Pick your tools

- Easiest: Python Flask + HTML + Bootstrap + MySQL
- Or Python Streamlit if you want the simplest option (almost no HTML needed)

Step 6: Connect Python to MySQL

- Install mysql-connector-python
- Write one file that connects to your database

Step 7: Build the pages (one by one)

1. Register and Login (Customer)
2. Add / View Projects
3. Add Rooms to a project
4. Add Materials
5. Generate Estimate
6. Make Payment
7. Assign Employees (admin side)

Each page does the same 4 things: Add, View, Edit, Delete.

Step 8: Test everything

- Go through the full flow: register, add project, add rooms, estimate, payment
- Fix errors

Step 9: Report and presentation

- ER diagram, schema, SQL queries, screenshots, conclusion

Where to start right now

Do Steps 1 to 3 first. Once your database works, the next task is the fixed SQL file and sample data.